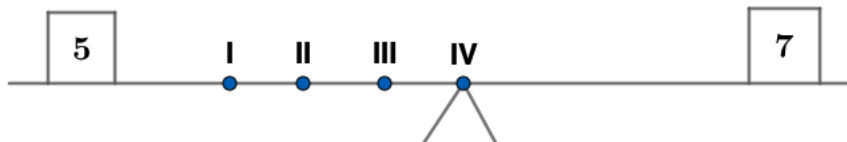
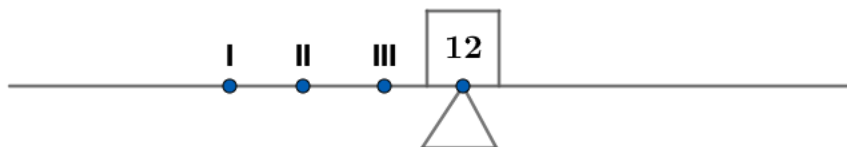


2022B F=ma Exam: Problem 3

Kevin S. Huang



Recall the torque produced by the weight of an extended object is equal to the torque produced by putting all the weight at the center of mass. Thus, the CM of the two blocks must be above the fulcrum at point IV for the seesaw to stay at rest. For the purposes of balancing torques, we can replace both blocks with a single block of mass $5\text{ kg} + 7\text{ kg} = 12\text{ kg}$ at the CM.



If the fulcrum is shifted to point II, then the object placed at point I has half the moment arm of the 12 kg block. Hence, it must be 24 kg so the answer is **C**.

